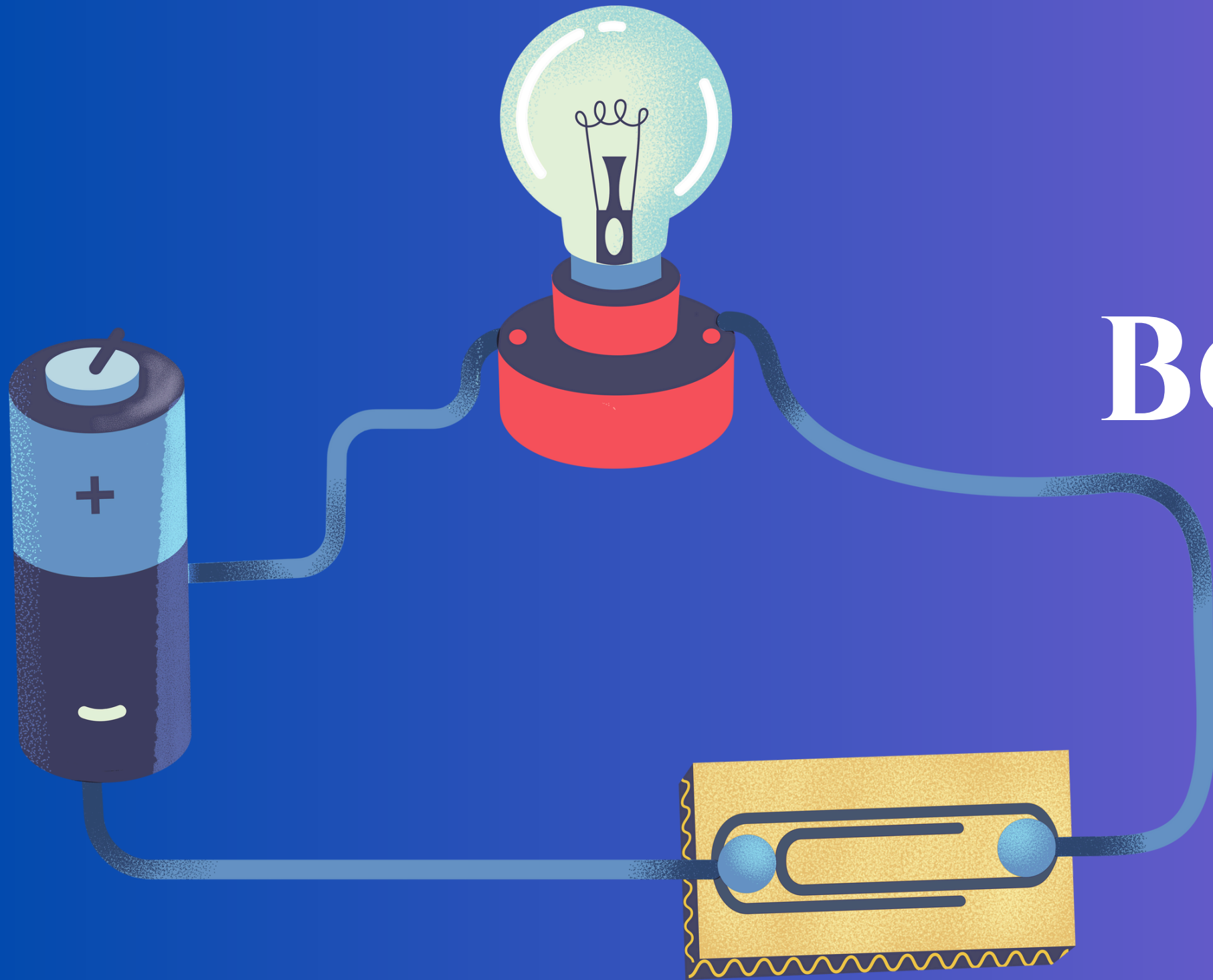




BOOLEAN ALGEBRA AND LOGIC CIRCUITS

CIRCUITS AND BOOLEAN ALGEBRA

- COMPUTERS ARE CONSTRUCTED FROM LOGIC CIRCUITS
 - LOGIC CIRCUITS PERFORM OPERATIONS ON DIGITAL SIGNALS
 - USUALLY IMPLEMENTED AS CIRCUITS WHERE SIGNAL VALUES ARE RESTRICTED TO A FEW VALUES

BINARY CIRCUITS

- CIRCUIT WITH 2 POSSIBLE STATES
 - OPEN ($X = 0$)
 - CLOSED ($X = 1$)
- X IS THE INPUT VARIABLE THAT CONTROLS THE SWITCH



BOOLEAN ALGEBRA

- NAME FOR GEORGE BOOLE, WHO INTRODUCED A SCHEME FOR DESCRIBING IN ALGEBRA THE PROCESSES INVOLVED IN LOGICAL REASONING
- BOOLEAN ALGEBRA WORKS WITH VARIABLES THAT CAN HAVE ONLY 2 VALUES - TRUE OR FALSE
- FUNDAMENTAL OPERATIONS
 - **AND**
 - **OR**
 - **NOT**

NOT OPERATION

- TAKES **ONE** VARIABLE/ARGUMENT/INPUT
- OUTPUT IS INVERSE OF INPUT
 - NOT(TRUE) = FALSE
 - NOT(FALSE) = TRUE

AND AND OR OPERATIONS

- **AND** FUNCTION OF VARIABLES $x_1, x_2, \dots, x_n$ HAS THE VALUE 1 ONLY IF ALL n VARIABLES ARE EQUAL TO 1.
 - LIKE MULTIPLICATION
- **OR** FUNCTION OF VARIABLES $x_1, x_2, \dots, x_n$ HAS THE VALUE 1 IF AT LEAST ONE, OR MORE, OF THE n VARIABLES ARE EQUAL TO 1
 - LIKE ADDITION

TRUTH TABLES

- USED TO DEFINE LOGIC OPERATIONS

x_1	x_2	$x_1 \cdot x_2$	$x_1 + x_2$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1
AND		OR	

TRUTH TABLE FOR AND AND
OR OPERATIONS WITH 2
INPUTS, x_1 AND x_2

EXTENDING AND AND OR

- AND AND OR CAN BE EXTENDED TO N VARIABLES

x_1	x_2	x_3	$x_1 \cdot x_2 \cdot x_3$	$x_1 + x_2 + x_3$
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

TRUTH TABLE FOR AND
AND OR OPERATIONS
WITH 3 INPUTS

AXIOMS OF BOOLEAN ALGEBRA

- B.A IS BASED ON AXIOMS
 - AXIOMS ARE A SET OF RULES THAT ARE DERIVED FROM A SMALL NUMBER OF BASIC ASSUMPTIONS
- BASIC ASSUMPTION - BOOLEAN ALGEBRA INVOLVES ELEMENTS THAT TAKE ON ONE OF TWO VALUES, 0 AND 1

AXIOMS OF BOOLEAN ALGEBRA

$$1a. \quad 0 \cdot 0 = 0$$

$$1b. \quad 1 + 1 = 1$$

$$2a. \quad 1 \cdot 1 = 1$$

$$2b. \quad 0 + 0 = 0$$

$$3a. \quad 0 \cdot 1 = 1 \cdot 0 = 0$$

$$3b. \quad 1 + 0 = 0 + 1 = 1$$

$$4a. \quad \text{If } x = 0, \text{ then } \bar{x} = 1$$

$$4b. \quad \text{If } x = 1, \text{ then } \bar{x} = 0$$

SINGLE VARIABLE RULES (THEOREMS)

$$5a. \quad x \cdot 0 = 0$$

$$5b. \quad x + 1 = 1$$

$$6a. \quad x \cdot 1 = x$$

$$6b. \quad x + 0 = x$$

$$7a. \quad x \cdot x = x$$

$$7b. \quad x + x = x$$

$$8a. \quad x \cdot \bar{x} = 0$$

$$8b. \quad x + \bar{x} = 1$$

$$9. \quad \bar{\bar{x}} = x$$

- FROM THE AXIOMS, WE DEFINE RULES (THEOREMS) FOR DEALING WITH SINGLE VARIABLES
- WE CAN PROVE BY PERFECT INDUCTION (SUBSTITUTE $x = 0$ AND $x = 1$ IN EACH EXPRESSION AND USE AXIOMS)

PRINCIPLE OF DUALITY IN B.A.

- GIVEN A LOGIC EXPRESSION, WE GET ITS DUAL BY REPLACING ALL $+$ WITH $\cdot$ (AND VICE VERSA) AND BY REPLACING 0S WITH 1S AND 1S WITH 0S.
- THE DUAL OF ANY TRUE STATEMENT, AXIOM OR THEOREM, IN BOOLEAN ALGEBRA IS ALSO A TRUE STATEMENT
- THIS MEANS THAT AT LEAST 2 DIFFERENT WAYS EXIST TO EXPRESS EVERY LOGIC FUNCTION

TWO- AND THREE- VARIABLE PROPERTIES

- WE OFTEN NEED TO DEAL WITH A NUMBER OF VARIABLES
- DEFINE SOME 2- AND 3-VARIABLE IDENTITIES
- GIVE THE DUAL OF EACH IDENTITY
- OFTEN REFERRED TO AS PROPERTIES
- IF x , y , AND z ARE VARIABLES IN B , THE FOLLOWING PROPERTIES HOLD...

PROPERTIES

$$10a. \quad x \cdot y = y \cdot x \qquad \text{Commutative}$$

$$10b. \quad x + y = y + x$$

$$11a. \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z \qquad \text{Associative}$$

$$11b. \quad x + (y + z) = (x + y) + z$$

$$12a. \quad x \cdot (y + z) = x \cdot y + x \cdot z \qquad \text{Distributive}$$

$$12b. \quad x + y \cdot z = (x + y) \cdot (x + z)$$

$$13a. \quad x + x \cdot y = x \qquad \text{Absorption}$$

$$13b. \quad x \cdot (x + y) = x$$

$$14a. \quad x \cdot y + x \cdot \bar{y} = x \qquad \text{Combining}$$

$$14b. \quad (x + y) \cdot (x + \bar{y}) = x$$

PROPERTIES

$$15a. \quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

DeMorgan's theorem

$$15b. \quad \overline{x + y} = \bar{x} \cdot \bar{y}$$

$$16a. \quad x + \bar{x} \cdot y = x + y$$

$$16b. \quad x \cdot (\bar{x} + y) = x \cdot y$$

$$17a. \quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

Consensus

$$17b. \quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

PROPERTIES

- WE CAN PROVE BY PERFECT INDUCTION (TRUTH TABLES), ALGEBRAIC MANIPULATION, OR VENN DIAGRAMS
- EXAMPLE: PROVING DEMORGAN'S THEOREM USING TRUTH TABLE

x	y	$x \cdot y$	$\overline{x \cdot y}$	$\bar{x}$	$\bar{y}$	$\bar{x} + \bar{y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0
		LHS		RHS		

VENN DIAGRAMS

- USING TRUTH TABLES TO VERIFY THEOREMS AND PROPERTIES CAN BE TEDIOUS, AND MAY NOT BE CONCEPTUALLY INFORMATIVE
- VENN DIAGRAMS ARE A TOOL THAT PROVIDES FOR MORE INTUITIVE UNDERSTANDING OF HOW EXPRESSIONS MAY BE EQUIVALENT
 - GRAPHICAL ILLUSTRATION OF SET ALGEBRA
 - WE'RE NOT GOING TO TALK ABOUT THEM.

EQUIVALENT NOTATIONS

The following notations are equivalent:

X' same as $\bar{X}$

$(X + Y)'$ same as $\overline{X + Y}$

OTHER NOTATION

- USING THE SAME SYMBOLS FOR ARITHMETIC AND LOGIC CAN BE CONFUSING
- WE CAN USE THE FOLLOWING NOTATION:

AND $\wedge$, *e.g.*, $x_1 \wedge x_2$

OR $\vee$, *e.g.* $x_1 \vee x_2$

PRECEDENCE OF OPERATIONS

- IN THE ABSENCE OF PARENTHESES, A LOGIC EXPRESSION IS EVALUATED IN THE FOLLOWING ORDER:

A. NOT

B. AND

C. OR

COMPUTERS AND ELECTRICITY

GATE

A DEVICE THAT PERFORMS A BASIC OPERATION ON ELECTRICAL SIGNALS

CIRCUITS

GATES COMBINED TO PERFORM MORE COMPLICATED TASKS

GATES

6 TYPES OF GATES

- NOT
- AND
- OR
- XOR
- NAND
- NOR

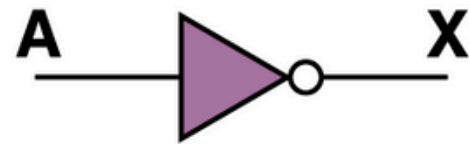
NOT GATE

A NOT GATE ACCEPTS ONE INPUT SIGNAL (0 OR 1)
AND RETURNS THE OPPOSITE SIGNAL AS OUTPUT

Boolean Expression

$$X = A'$$

Logic Diagram Symbol



Truth Table

A	X
0	1
1	0

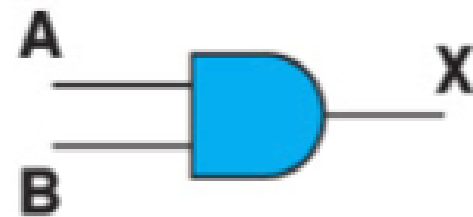
AND GATE

AN AND GATE ACCEPTS TWO INPUT SIGNALS
IF BOTH ARE 1, THE OUTPUT IS 1;
OTHERWISE, THE OUTPUT IS 0

Boolean Expression

$$X = A \cdot B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

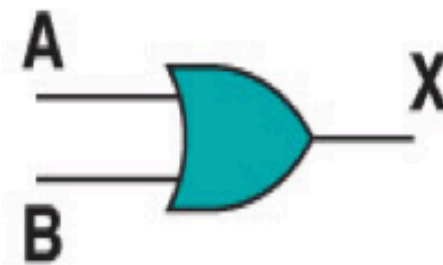
OR GATE

AN OR GATE ACCEPTS TWO INPUT SIGNALS
IF BOTH ARE 0, THE OUTPUT IS 0; OTHERWISE,
THE OUTPUT IS 1

Boolean Expression

$$X = A + B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

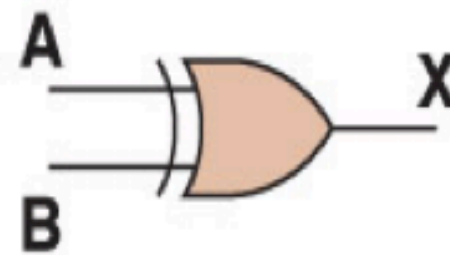
XOR GATE

AN XOR GATE ACCEPTS TWO INPUT SIGNALS
IF BOTH ARE THE SAME, THE OUTPUT IS 0;
OTHERWISE, THE OUTPUT IS 1

Boolean Expression

$$X = A \oplus B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

XOR GATE CONT...

- NOTE THE DIFFERENCE BETWEEN THE XOR GATE AND THE OR GATE; THEY DIFFER ONLY IN ONE INPUT SITUATION
- WHEN BOTH INPUT SIGNALS ARE 1, THE OR GATE PRODUCES A 1 AND THE XOR PRODUCES A 0
- **XOR** IS CALLED THE EXCLUSIVE OR
- **OR** IS THE INCLUSIVE OR

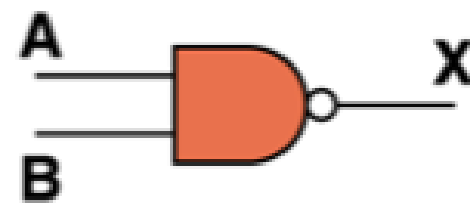
NAND GATE

THE NAND GATE ACCEPTS TWO INPUT SIGNALS
IF BOTH ARE 1, THE OUTPUT IS 0; OTHERWISE, THE
OUTPUT IS 1

Boolean Expression

$$X = (A \cdot B)'$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

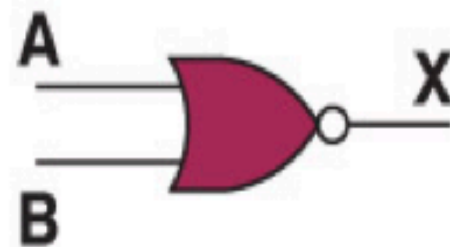
NOR GATE

THE NOR GATE ACCEPTS TWO INPUT SIGNALS
IF BOTH ARE 0, THE OUTPUT IS 1; OTHERWISE, THE
OUTPUT IS 0

Boolean Expression

$$X = (A + B)'$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

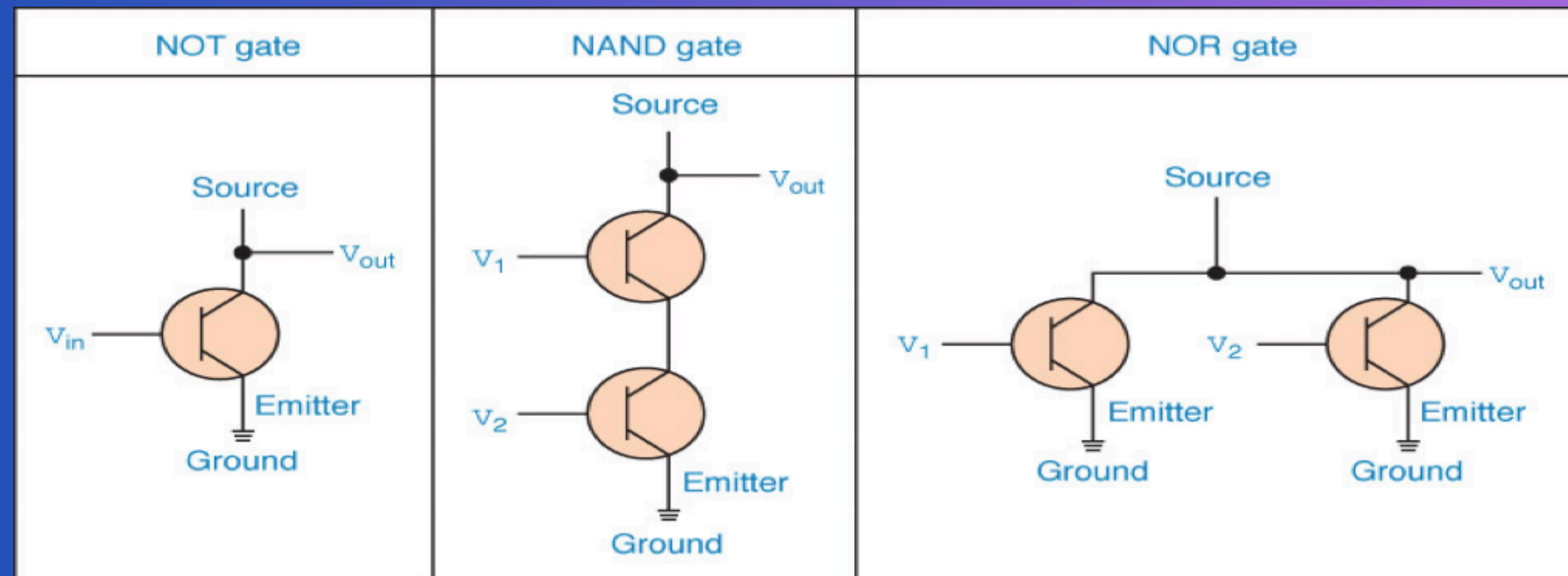
CONNECTING GATES

- GATES ARE CONSTRUCTED WITH **TRANSISTORS**

(TRANSISTOR COUNT: [HTTP://EN.WIKIPEDIA.ORG/WIKI/TRANSISTOR_COUNT](http://en.wikipedia.org/wiki/Transistor_Count))

- A DEVICE THAT ACTS EITHER AS A WIRE THAT CONDUCTS ELECTRICITY OR AS A RESISTOR THAT BLOCKS THE FLOW OF ELECTRICITY, DEPENDING ON THE VOLTAGE LEVEL OF AN INPUT SIGNAL

- THE EASIEST GATES TO CREATE ARE THE NOT, NAND, AND NOR GATES
- YOU WILL LEARN MORE ABOUT THIS IN YOUR DIGITAL SYSTEMS COURSE

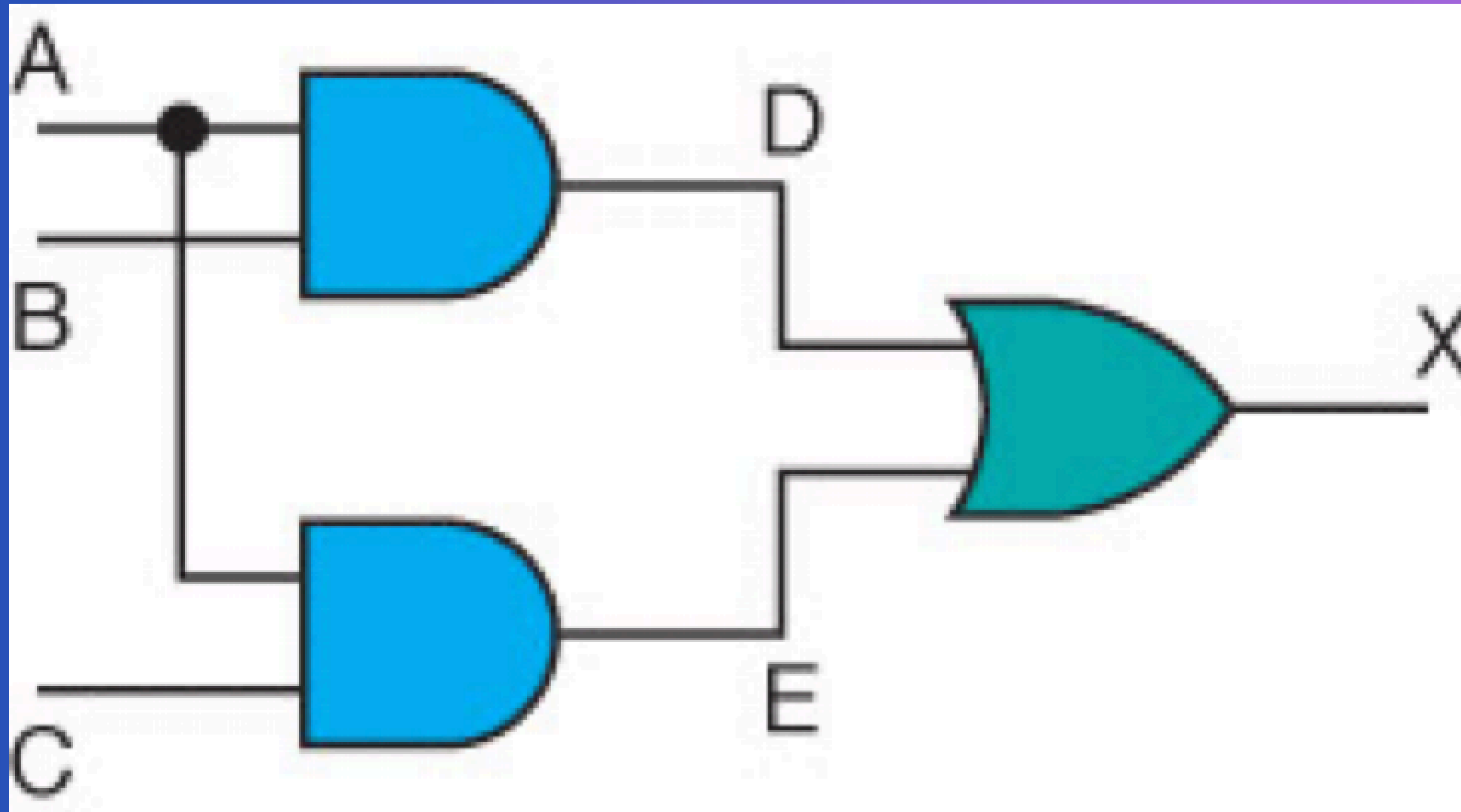


CIRCUITS

- **COMBINATIONAL CIRCUIT**
 - THE INPUT VALUES EXPLICITLY DETERMINE THE OUTPUT
- **SEQUENTIAL CIRCUIT**
 - THE OUTPUT IS A FUNCTION OF THE INPUT VALUES AND THE EXISTING STATE OF THE CIRCUIT
 - WE DESCRIBE THE CIRCUIT OPERATIONS USING
 - BOOLEAN EXPRESSIONS
 - LOGIC DIAGRAMS
 - TRUTH TABLES

COMBINATIONAL CIRCUITS

GATES ARE COMBINED INTO CIRCUITS BY USING THE OUTPUT OF ONE GATE AS THE INPUT FOR ANOTHER



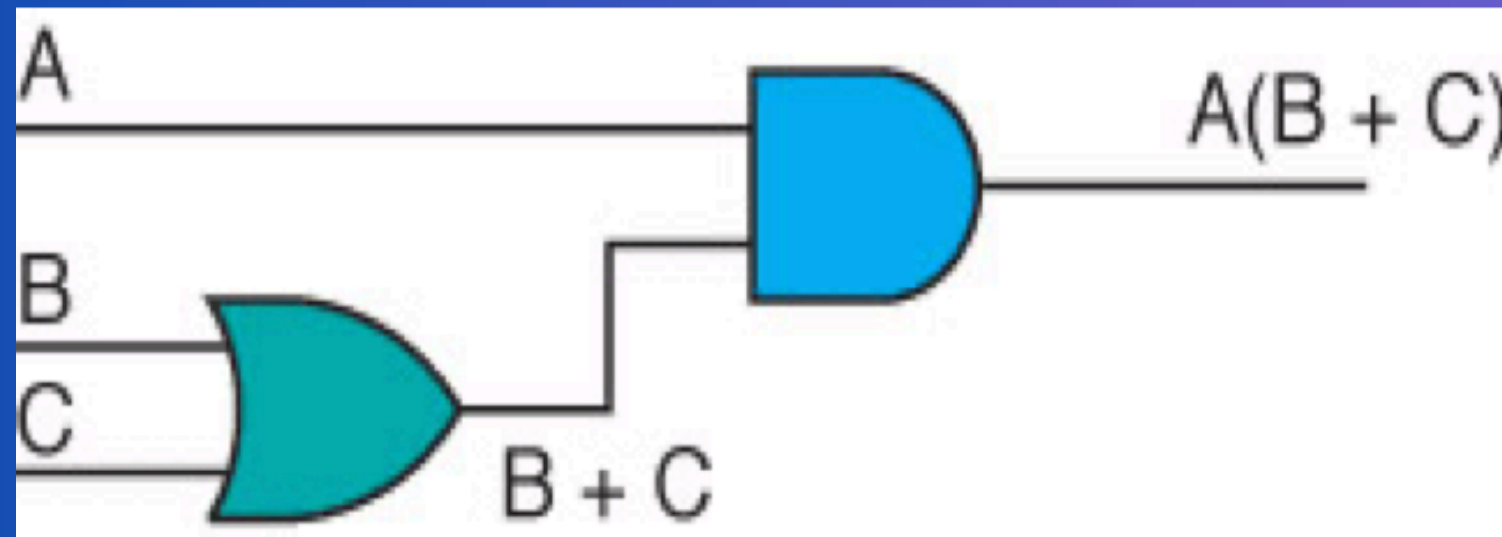
COMBINATIONAL CIRCUITS

A	B	C	D	E	X
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

- THREE INPUTS REQUIRE EIGHT ROWS TO DESCRIBE ALL POSSIBLE INPUT COMBINATIONS
- THIS CIRCUIT IS DESCRIBED BY THE BOOLEAN EXPRESSION $(AB + AC)$

COMBINATIONAL CIRCUITS

CONSIDER THE FOLLOWING BOOLEAN EXPRESSION $A(B + C)$



A	B	C	$B + C$	$A(B + C)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

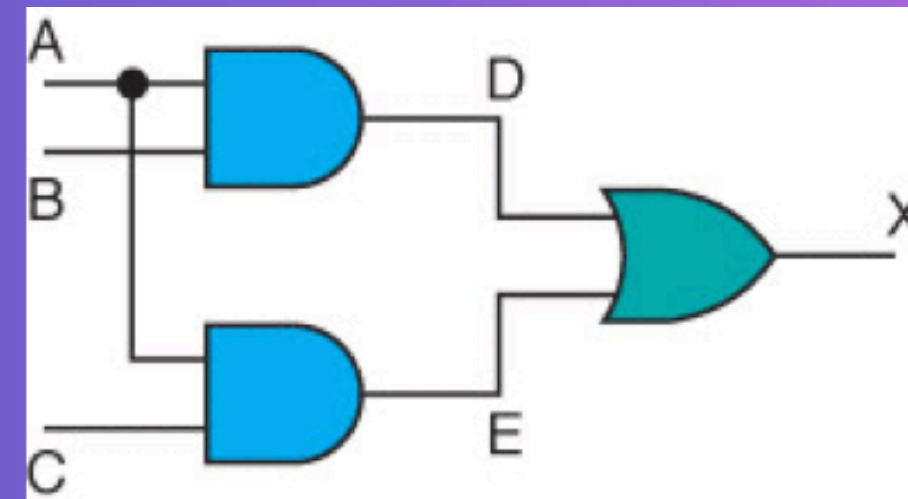
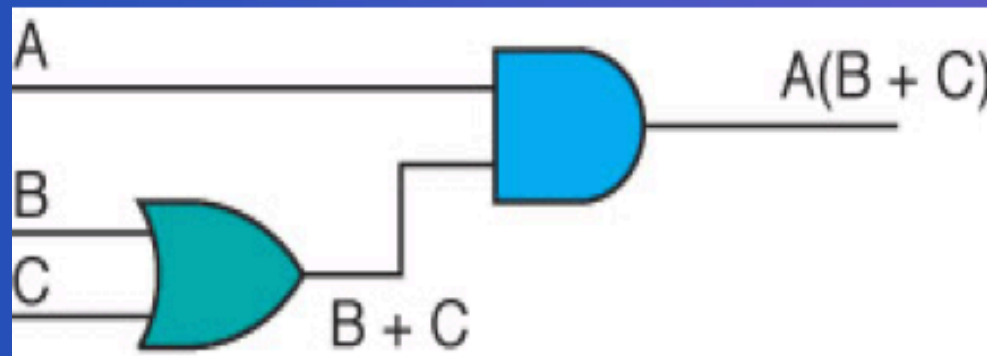
DOES THIS TRUTH TABLE LOOK FAMILIAR?

COMPARE IT WITH PREVIOUS TABLE...

COMBINATIONAL CIRCUITS

CIRCUIT EQUIVALENCE

- TWO CIRCUITS THAT PRODUCE THE SAME OUTPUT FOR IDENTICAL INPUT
- BOOLEAN ALGEBRA ALLOWS US TO APPLY PROVABLE MATHEMATICAL PRINCIPLES TO HELP DESIGN CIRCUITS
- $A(B + C) = AB + AC$ (DISTRIBUTIVE LAW) SO CIRCUITS MUST BE EQUIVALENT



ADDERS

- AT THE DIGITAL LOGIC LEVEL, ADDITION IS PERFORMED IN BINARY
- ADDITION OPERATIONS ARE CARRIED OUT BY SPECIAL CIRCUITS CALLED, APPROPRIATELY, **ADDERS**

ADDERS

THE RESULT OF ADDING
TWO BINARY DIGITS COULD
PRODUCE A CARRY VALUE
RECALL THAT $1 + 1 = 10$
IN BASE TWO

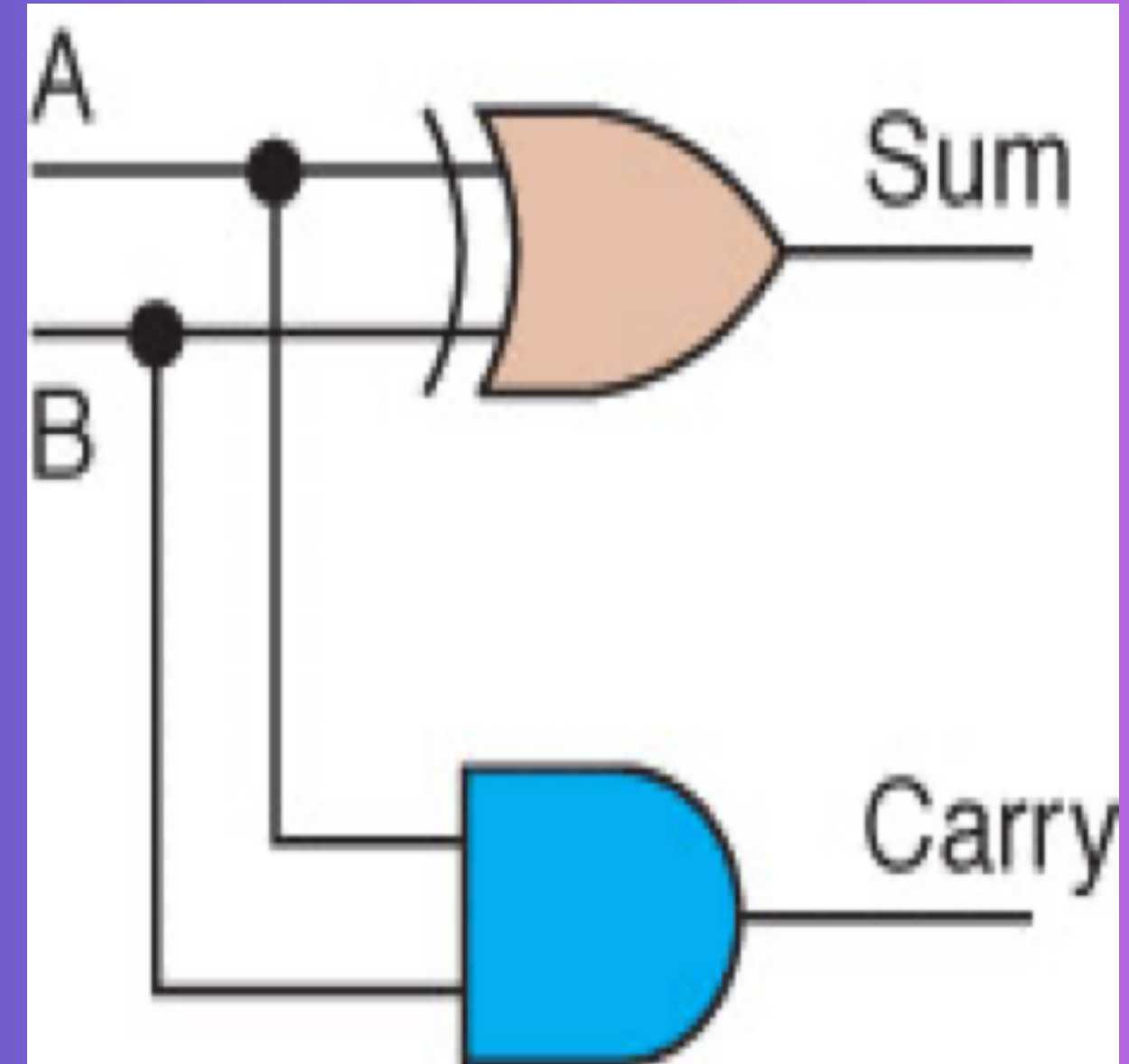
HALF ADDER

A CIRCUIT THAT COMPUTES
THE SUM OF TWO BITS
AND PRODUCES THE
CORRECT CARRY BIT

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

ADDERS

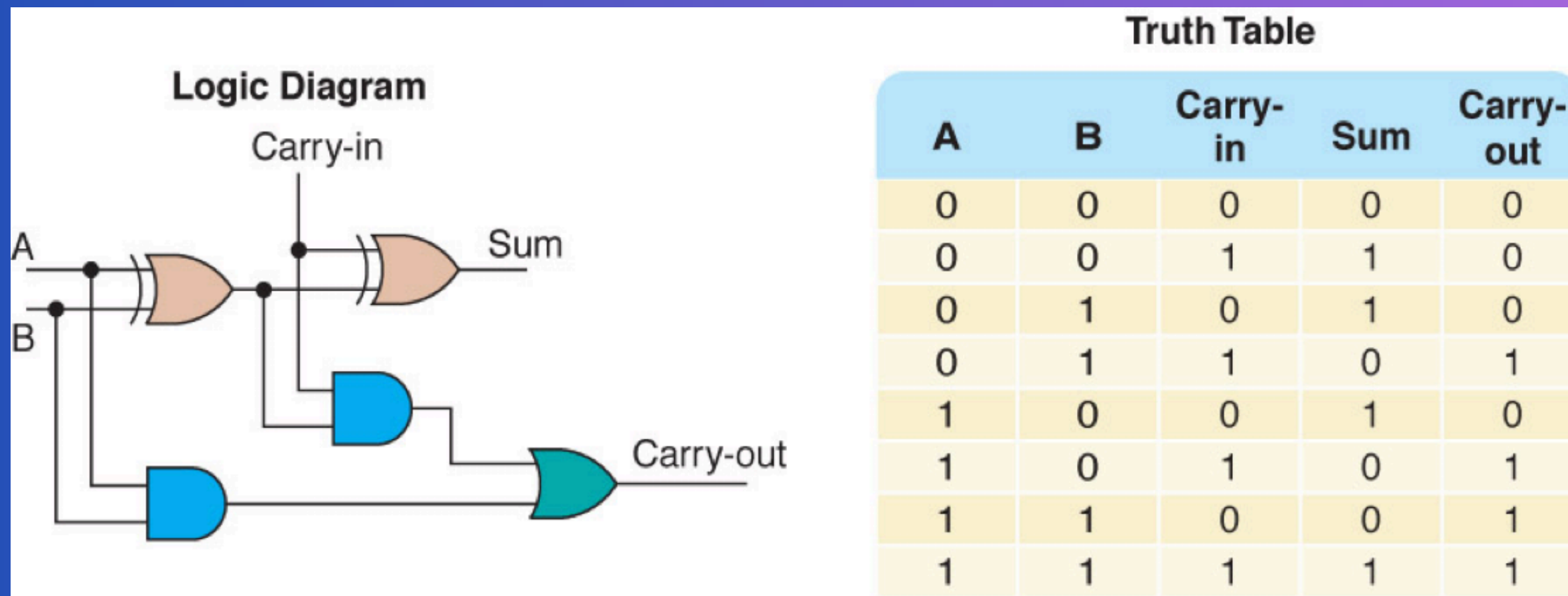
- CIRCUIT DIAGRAM REPRESENTING A HALF ADDER
- BOOLEAN EXPRESSIONS
 - $SUM = A \oplus B$
 - $CARRY = AB$



ADDERS

FULL ADDER

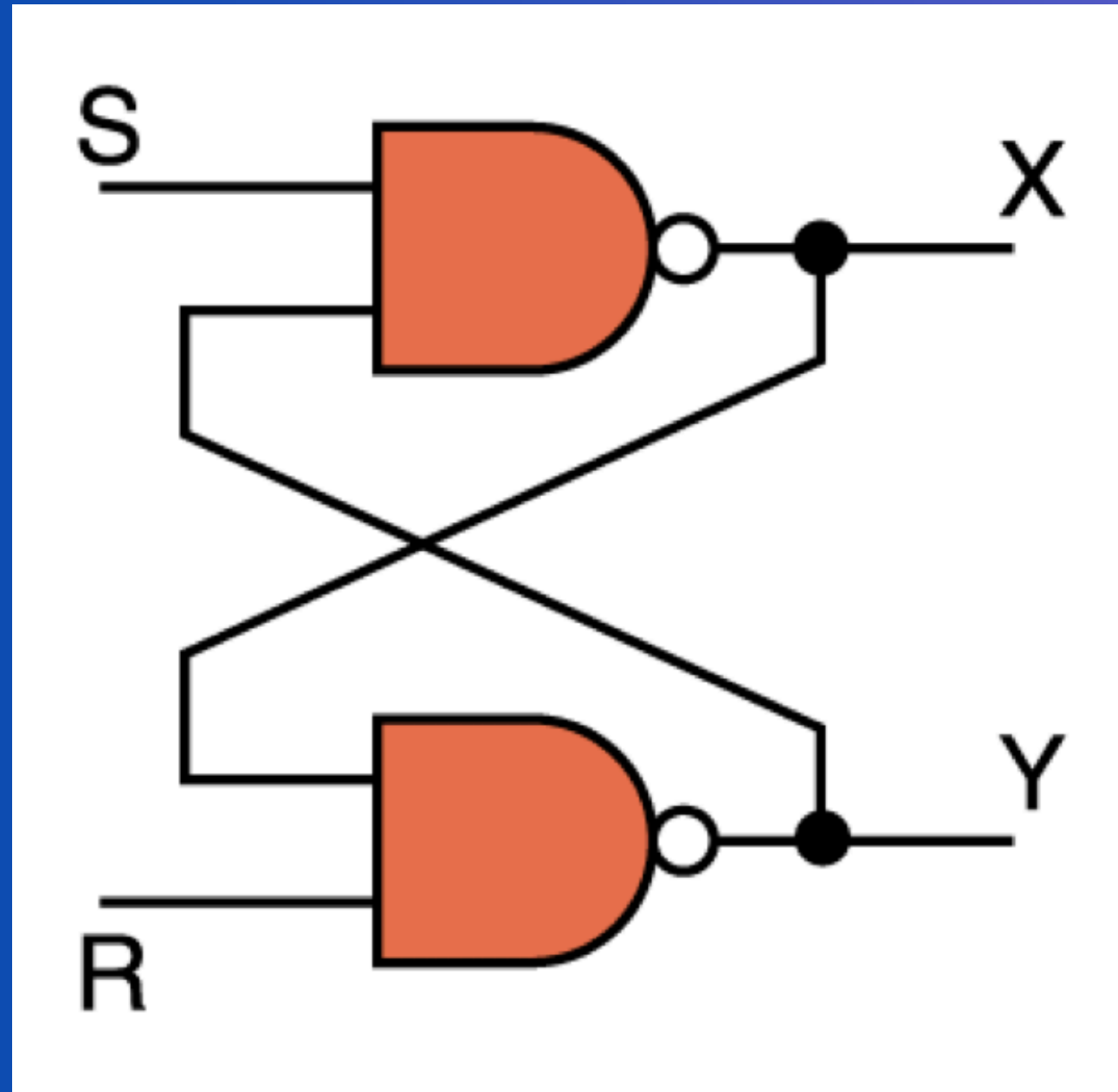
A CIRCUIT THAT TAKES THE CARRY-IN VALUE INTO ACCOUNT



CIRCUITS AS MEMORY

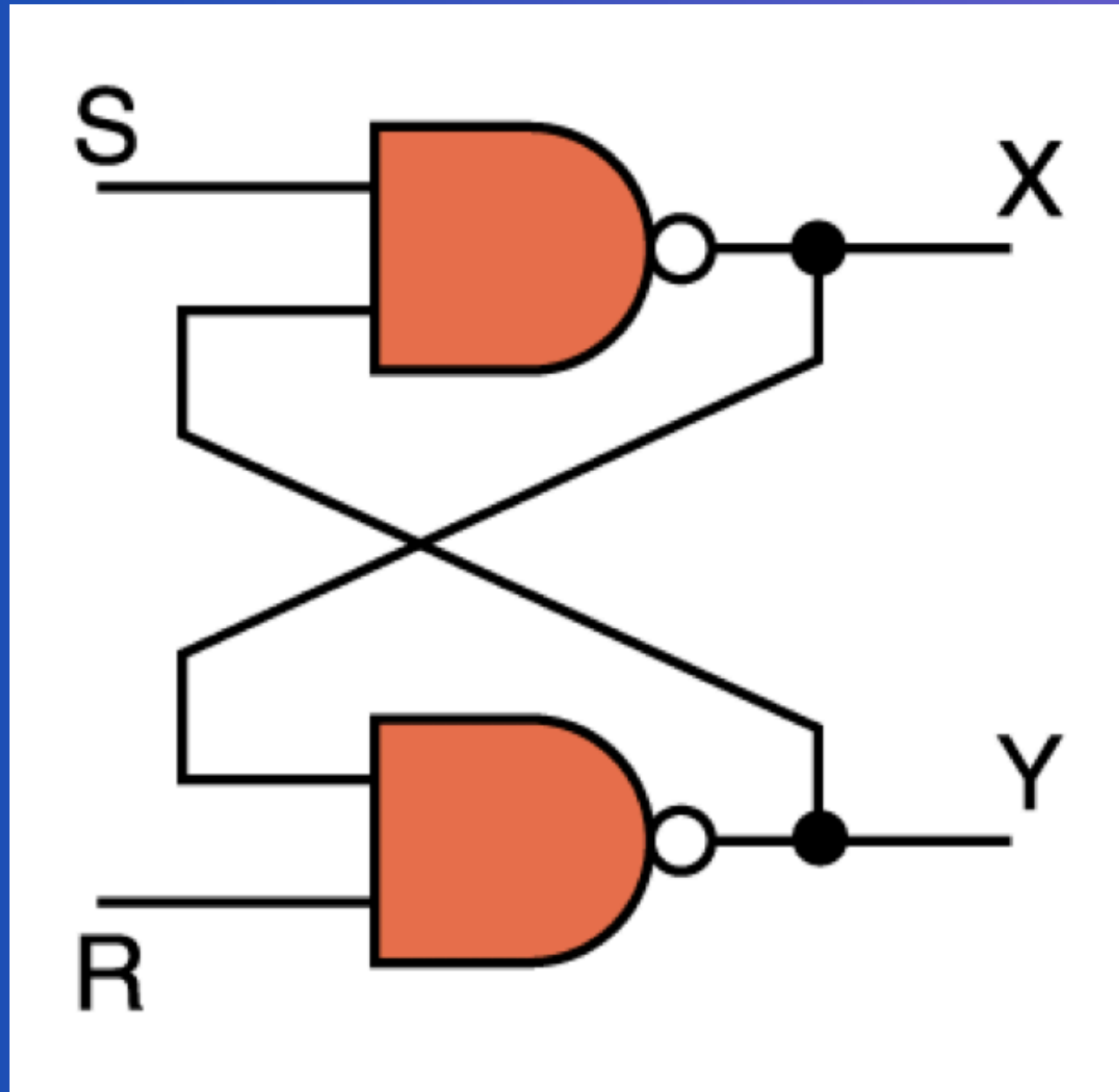
- DIGITAL CIRCUITS CAN BE USED TO STORE INFORMATION
- THESE CIRCUITS FORM A SEQUENTIAL CIRCUIT, BECAUSE THE OUTPUT OF THE CIRCUIT IS ALSO USED AS INPUT TO THE CIRCUIT

CIRCUITS AS MEMORY



- AN S-R LATCH STORES A SINGLE BINARY DIGIT (1 OR 0)
- THE OUTPUT OF ONE GATE IS USED AS AN INPUT TO THE OTHER GATE

CIRCUITS AS MEMORY



- THE DESIGN OF THIS CIRCUIT GUARANTEES THAT THE TWO OUTPUTS X AND Y ARE ALWAYS COMPLEMENTS OF EACH OTHER
- THE VALUE OF X IS CONSIDERED TO BE THE CURRENT STATE OF THE CIRCUIT
 - IF X IS 1, THE CIRCUIT IS STORING A 1
 - IF X IS 0, THE CIRCUIT IS STORING A 0

CIRCUITS AS MEMORY

S-R LATCH TRUTH TABLE

S	R	X	Y
0	0	Not allowed (forbidden state)	
0	1	1	0
1	0	0	1
1	1	No change in the output	

INTEGRATED CIRCUITS

INTEGRATED CIRCUIT (ALSO CALLED A **CHIP**)

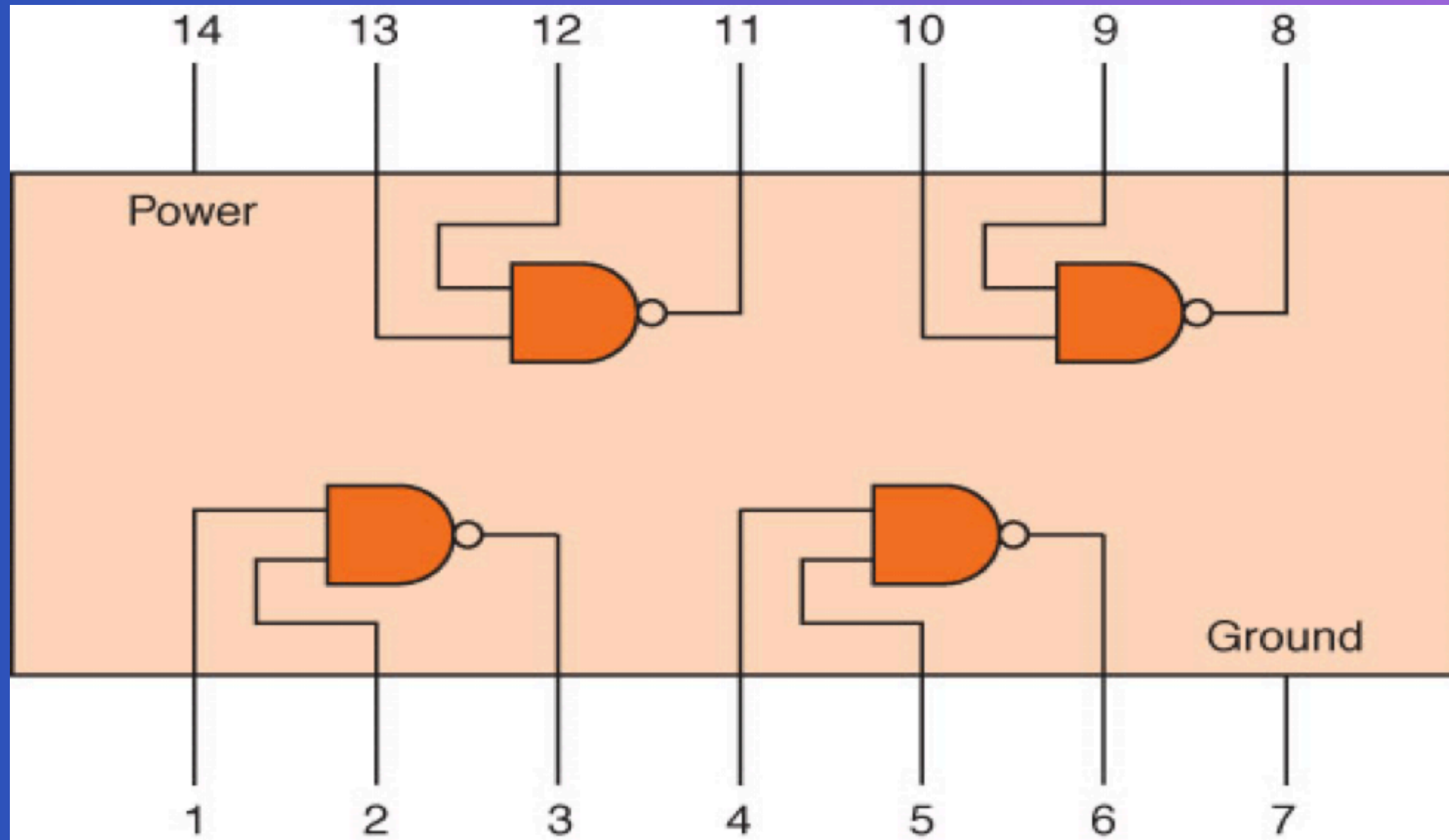
- A PIECE OF SILICON ON WHICH MULTIPLE GATES HAVE BEEN EMBEDDED
 - SILICON PIECES ARE MOUNTED ON A PLASTIC OR CERAMIC PACKAGE WITH PINS ALONG THE EDGES
 - PINS CAN BE SOLDERED ONTO CIRCUIT BOARDS OR INSERTED INTO APPROPRIATE SOCKETS

INTEGRATED CIRCUITS

INTEGRATED CIRCUITS (IC) ARE CLASSIFIED BY THE NUMBER OF GATES CONTAINED IN THEM

Abbreviation	Name	Number of Gates
SSI	Small-Scale Integration	1 to 10
MSI	Medium-Scale Integration	10 to 100
LSI	Large-Scale Integration	100 to 100,000
VLSI	Very-Large-Scale Integration	more than 100,000

INTEGRATED CIRCUITS



EXAMPLE SSI CHIP CONTAINING INDEPENDENT NAND GATES

CPU CHIPS

- THE MOST IMPORTANT INTEGRATED CIRCUIT IN ANY COMPUTER IS THE CENTRAL PROCESSING UNIT, OR CPU
- EACH CPU CHIP HAS A LARGE NUMBER OF PINS THROUGH WHICH ESSENTIALLY ALL COMMUNICATION IN A COMPUTER SYSTEM OCCURS
- WE WILL LOOK AT THE CPU IN MORE DETAIL SOON